Trinh Gia Hiep - 2352343 - HW4

1. Let $\theta = P(\text{head})$. With 9 heads & 3 tails in 12 tosses.

$L(\theta) \propto \theta^9 (1-\theta)^3, \quad 0 \le \theta \le 1$

- How the data shapes plausibility:
- $\theta = 0, \theta = 1$ are ruled out.
- The likelihood peaks at $\theta = 9/12 = 0,75$.
- Plausibility falls off smoothly away from the peak.

2. Counts (8 examples):

	Y = 0	Y = 1
Count	4	4
A = 1	1	3
B = 0	2	2

Priors $P(Y=0) = 4/8 = 0,5$

$P(Y=1) = 0,5$

Add 1 smoothing: $(\text{count} + 1) / N_y + 2 = 6$

Probability	Value
$P(A=1 \mid Y=0)$	$(1+1)/6 = 1/3$
$P(B=0 \mid Y=0)$	$(2+1)/6 = 1/2$
$P(A=1 \mid Y=1)$	$(3+1)/6 = 2/3$
$P(B=0 \mid Y=1)$	$(2+1)/6 = 1/2$

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Scores for $A=1, B=0$:

- $Y=0: 0,5 \cdot 1/3 \cdot 1/2 = 1/12 \approx 0,0833$
- $Y=1: 0,5 \cdot 2/3 \cdot 1/2 = 1/6 \approx 0,1667$

Normalized: $P(Y=1 \mid A=1, B=0) = \frac{0,1667}{0,0833+0,1667} = \frac{2}{3}$

$$P(Y=0 \mid A=1, B=0) = \frac{1}{3}$$

Prediction: $Y=1$

Unrealistic assumption: Naive Bayes assume $A \perp B \mid Y$, In real data, attributes are often correlated even within a class.

3. DAG: $A \to C$, $B \to C$, $C \to D$, $B \to E$

C is a collider on $A \to C \leftarrow B$

Pair	Evidence	Path	Set Status	Justification
(i) A, B	none	$A \to C \leftarrow B$	Blocked	the collider C
(ii) A, B	C	$A \to C \leftarrow B$	Active	
(iii) A, D	C	$A \to C \to D$	Blocked	
(iv) C, E	B	$C \leftarrow B \to E$	Blocked	

Justification: (i) the collider C is unobsered, also descendant D $\Rightarrow$ the path is close. $A \perp B$

(ii): Observing the collider opens the path $\Rightarrow$ A & B dependent.

(iii) C is a chain node, and is observed, so it blocks the path. $A \perp D \mid C$

(iv): B is a fork node, " " " $C \perp E \mid B$.

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No.

Date

4. Evidence changes beliefs.

- Predictive reasoning (cause → Effect):

. Evidence: Dark clouds and a rain forest.

. Inferred: The streets will likely be wet.

- Diagnostic reasoning (effect → cause).

. Evidence: The car won't start and the headlights are dim.

. Inferred: The battery is dead.

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